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Exploratory Study of Artificial Intelligence Techniques for Detecting Milk Adulteration in Food Safety

Sulochana V^{1,*}, S. Prabhu² and Basheer Riskhan³

¹Associate Professor, Department of Computer Science, Hindusthan College of Arts and Science, Coimbatore - 641028, India.

²Assistant Professor, Department of Computer Science, KG College of Arts and Science, Coimbatore - 641035, India.

³Dean and Associate Professor, School of Computing and Informatics, Albukhary International University, Malaysia.

*Corresponding Author: Sulochana V. Email: sulochana.v@hicas.ac.in

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ABSTRACT: Food adulteration is the process where the quality is compromised by adding certain chemicals or other substitutions, which, when consumed, cause health hazards. The process includes not only the purposeful addition of additives but also the contamination occurring in the growing stage, storage, and when distributed. Adulteration has become a big business. We belong to a land where our ancestors taught that “food is medicine. Milk is one of the important foods consumed irrespective of age that ranging from infants to old people. In contrast, in the recent business era, milk is getting adulterated in more complicated ways that bypass the normal tests, and hence, there is a huge demand for a cutting-edge technology for detecting the same. The application of Artificial Intelligence has emerged in many diverse fields, not limited to the identification of food adulteration. This paper gives a gist of the most significant research that has been carried on in the detection of adulterated food, particularly milk, using the techniques of machine learning. Special attention is kept on the Machine Learning techniques that are designed specifically to detect milk adulteration.

KEYWORDS: Adulteration, Artificial Intelligence, Machine Learning, Contamination.

1 Introduction

The adulteration of milk and other dairy products has grabbed the attention globally after the identification of melamine particles in milk in Chinese products in the year 2018 [1]. The practice of milk adulteration is a very old scenario. The milk scandal that was reported in the year 1850 caused nearly 8000 infants to lose their lives in the city of New York. Milk is one of the chiefs and most used food items and has nutrients required for all ages of humans. It is also a rich source of many essential minerals and proteins. Milk is subject to adulteration easily across the globe. The prime reasons behind this are the gap between supply and demand and high business motives. This also occurs due to the affordability of the customers to buy and the lack of efficient mechanisms to detect the same [2]. Although the motive is economic, the impact the adulteration costs lives [3]. The situation is unmanageable in most of the underdeveloped and developing countries owing to the lack of efficient mechanisms and proper laws to enforce strict action against the culprits. Adulteration of milk can be identified either using qualitative or quantitative mechanisms. The former uses various chemical reactions for the purpose, and the latter identifies the nature of the adulterants. Milk adulteration



detection methods need a swift transition from old manual methods to highly accurate machine-based techniques [4].

Food Safety policies and tools have been generated and are being it in all over the world. On adopting the available tools, still, many challenges still arise in various aspects, such as the accuracy of contamination prediction. The issue of food adulteration is a contemporary mechanism, and it is as old as the food production techniques themselves. Owing to the lower accuracy and time complexity involved in the conventional methods of identifying milk adulterity, modern scientific development is needed, and since there are a large variety and complex information available on this, many data mining techniques in terms of supervised [5] and unsupervised models have been employed. Those models focus on the microbiological or toxic chemical hazards. The ultimate goal of the study is to examine the data mining technique employed for food adulteration and contamination. The remaining sections of the paper are organized as follows: Section 2 gives the preliminaries on the types of machine learning, and Section 3 gives the best research works carried out on each of them, with the methods, advantages, and the area for improvement. Section 4 gives a comparative study on the results obtained through the methods adopted in the literature, and the conclusion and future directions are projected in Section 5.

2 Preliminaries

2.1 Machine Learning

Machine Learning is the buzzword in the current digital era. The important reason behind this is the increase in data accumulation and a parallel increase in computational powers and hardware capacities. ML is almost used everywhere, from automation to producing intelligent decisions in almost every domain [6]. The ML methods have hit the floor in such a way that the users themselves would have already been using the devices without their understanding. ML techniques are those that enable the system to learn from the Data and improve its performance itself without being given explicit programming. ML is a category of algorithms that allows the application software to be more accurate in the process of outcome prediction without additional programming [7]. The basic intent of ML methods is to constitute algorithms that can get data as input and use various statistical methods for predicting the output as and when new data are accumulated.

2.2 Types

Machine Learning algorithms are more generally classified as supervised, unsupervised, and reinforcement learning [8].

In the case of supervised methods, the system is given the data along with the labels, which indicate that all the data are tagged correctly with the label. The motive is to make the mapping function appropriate such that when we have X as the input data, the output Y can be predicted. The important methods of supervised learning are i. Classification where the output variable is of any of the categories, such as "color", "shape", or simple YES or NO, and the other methods are the regression models.

As the unsupervised learning is considered, the Artificial Intelligence systems here are presented with data that are not labelled and which are not categorized, and the algorithms act upon the data without training. The output here is dependent on the algorithms, which are pre-coded. Using unsupervised learning is a kind of testing of the AI systems. The unsupervised learning is normally classified as i. clustering, where the intention is to discover under which group a particular entity falls, and ii. An association where the intention is to identify the rules that describe a large part of the data, such that "customers who bought X are likely to buy Y also".

A reinforcement learning method is where the agent interacts with the environment for learning purposes. The agents here get the rewards if they perform correctly and penalties if they malfunction. The agent here learns without the influence of the human for maximum rewards and minimum penalty [9]. This is also considered a dynamic programming, which trains the algorithms to use a system of rewards and penalties. The commonly used methods are i. Q-Learning, which is considered to be an offshore policy learning, where the algorithm seeks to identify the best-fit action to be performed if the current state is given. It is termed offshore because the systems' actions are outside the present policy, such that random actions. ii. Deep learning, where the ML techniques use multiple level of layers for progress through high-level features from the input. For instance, in the image processing domain, the low level may identify edges and the high levels identify the core concepts that are relevant to humans, such as digits or letters. Figure 1 gives a generalized view of the classification of ML techniques.

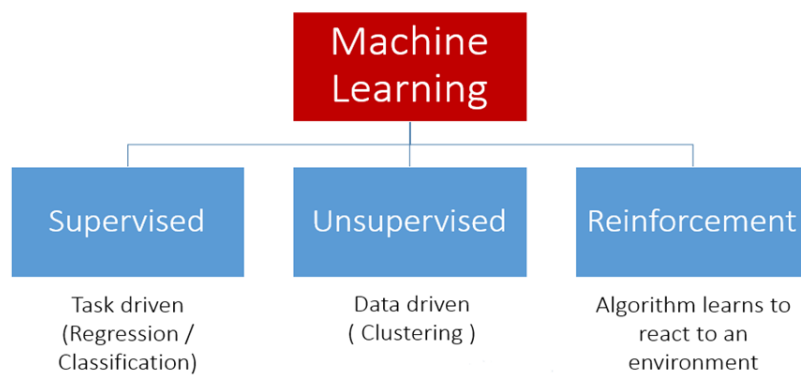


Figure 1: Generic classification of ML methods

3 Studies on ML methods in food adulteration

3.1 Review of supervised learning in food adulteration detection

Wabben et al [10] proved that the concurrent use of both targeted and untargeted models, which are applied to the FT-IR spectrum in identifying milk adultery, is efficient. By the use of appropriate methods, the detection limits, sensitivity, and specificity are identified. The proposed model used a multivariate linear regression model for the practical implications of the method. The theoretical limitations are also provided for a potential range of adulterants used. The performance metrics used here are the Precision, F-score, and Recall.

Zhang et al [11] proposed a series of samples of adulterated milk by adding different levels of aqueous dextrin. The improvised Support Vector Machines are used for the purpose of building the classification model for differentiating the milk that was adulterated based on the NIR spectrum of various sample sets. Uniformity in the design of tables was made use of. The results showed that the milks in which the adulterated content is greater than 5 % are well detected by the proposed SVM combined with NIR based spectroscopy, with a very high validation ratio. The proposed methods outperformed other methods in terms of Specificity, Sensitivity, and Correlation values.

Neto et al (2019) introduced the ML algorithms on milk spatial data. The proposed methods were able to find the most common adulteration methods that prevail in the dairy industry. Both classification and linear regression methods were used for the binary and multiclass means of classification, and the experimental results were compared. The proposed classification and regression methods were able to perform better than the FITR device and also the other traditional

methods of dairy manufacturing. The performance measures included the cosine value and the correlation.

Mahood et al (2017) developed a new infrared-based spectroscopy (NIR), which is combined with that of the multivariate linear regression for identifying the level of formalin in cows' milk. The methods, such as PCA and partial test regression (PCA), are applied for a statistical investigation of the NIR data. The PLS-NIR methods were used for checking the difference between the pure and adulterated milk. A regression model using PLS was built for quantifying the formalin levels in milk. The proposed method used Entropy, F Score, Precision, and recall as the performance measures.

Rebechi et al (2016) conducted research to test for accuracy in the fatty acid for the detection of adulterants, which replaced milk fat with that of animal fat, and also proposed a regression model for the evaluation of this kind of adulterant. On every adulterant, the Multiple Linear model of regression (MLR) was implied by choosing a model, and later it was validated. The matrix between the calibrations and validation is constructed by implementing a genuine mechanism to differentiate between pure and adulterated milk. The proposed models were able to find the milk adulteration at levels which are greater than 10%. The proposed method was better in terms of Specificity, Sensitivity, and Entropy when compared to other techniques.

Table 1 shows the tabular description of the most important research carried out using supervised ML methods for milk adulteration detection.

Table 1: Review of supervised methods

Author(s)	Method	Advantages	Metrics
Wabben et al	FT-IR Spectra	Identification of Potential Adulterants	F score, Precision, Recall
Zhang et al	Improved v-support vector machines	Accuracy increased to 5%	Specificity and Sensitivity, Correlation
Neto et al	Principle component analysis (PCA)	Outperforms traditional FITR	Cosine similarity and correlation
Mahood et al	Partial least discriminant analysis	Identifies Formalin Adulterant	Entropy, F Score, Precision, and Recall
Rebechi et al	Multiple Linear Regression	Accuracy increase over 10%	Specificity, Sensitivity, Entropy

Roney et al (2015) investigated the effect of Raman Microscopy and the ANN as a practical means of assessing fluid milk by the addition of whey to it. The Quantitative findings of adulteration were done through the feed-forward ANN. Various configurations of the Neural Networks were made and are evaluated based on their coefficient. The proposed model observed that the data from both training and validation had a high value for Recall, such as 99.9% which proved that the combined method of ANN and the Raman spectroscopy is a very fast, simple, and efficient method for the identification of quality milk from being adulterated by whey. The model outperformed others in terms of Root Mean Square and Average area under the curve.

Jiang et al (2019) made an attempt to develop an integrated mechanism that used least-square SVMs with the NMR technique for the identification of the level of adulteration in milk powder. It is proven that the proposed model can better identify adulterated milk. The experimental results prove

that the proposed SVM-NMR can yield very high and dense clusters with an accuracy of more than 98%. The proposed method used F Score, Precision, and recall as its performance metrics.

Hsieh et al (2011) implemented the LS-SVM method for the calibration and also a prediction model to identify adulteration in the cows' milk. The grid-based search technique was used to arrive at a better value for the Network parameters. The graph from the results shows that the adulteration ratio of above 10% is very well distinguished from those that are pure. The proposed method uses F-score, Precision, and Recall as its performance metrics.

Kuswandi et al (2015) made use of a couple of datasets that were recorded using the FTIR means of Spectroscopy using the total attenuated reflection (ATR) based on the spectrum data. The chemical analysis was made using the various modelling methods, such as LDA, SMCA, and SVM, on the data. At last, the optimum results of the models in the data set on the analysis of milk adulteration were selected, and the best-fit model was compared with the famous ELISA method. The proposed method seemed more efficient than that of the ELISA model in terms of Specificity, Sensitivity, and Correlation.

Leonardi et al (2015) discussed the ML method, which aims the define in order to assess the authenticity of some high-value milk powders. It was also demonstrated that the classification's effectiveness is very high in the proposed model in terms of the ability to exploit features on the chemical profile of milk that are obtained from biochemical methods. The proposed method was found to be more efficient when applied to real-time data and with synthetic data generated from a Bayesian generative model.

Table 2 shows the gist of research carried out using unsupervised learning for milk adultery identification with the metrics used.

Table 2: Review of unsupervised methods

Author(s)	Method	Advantages	Metrics
Roney et al	Artificial neural network	Simple and efficient	Root mean square and Average area under the curve
Jiang et al	Least-squares support vector machine	Accuracy is 98%	F score, Precision, Recall
Hsieh et al	Grid search SVM	Accuracy increases by 10%	Specificity and Sensitivity, Correlation
Kuswandi et al	Fourier Transform Infrared (FTIR)	Better than the ELISA method	Cosine similarity and correlation
Leonardi et al	Bayesian network model	Identifies Chemical ingredients in a better way	Entropy, F Score, Precision, and Recall

3.2 Review on reinforcement learning for food adulteration

Lei-Hong et al (2019) provided a solution for the automatic identification of common beetle species that frequently contaminate milk products. The approach was based on the convolutional neural network, which trained a dataset having 6900 images of various elytra fragments. The proposed methods acquired an accuracy level of 83% when cross-validated. It is also to be noted that the performance of the classification is obtained without the need to design and select the images that are domain specific, which depicts the deep learning importance in milk adultery. The proposed methods produced better results in terms of Root mean square and Average area under the curve.

Al-Sarayreh et al (2018) investigated the accuracy of hyperspectral imaging for the authentication and classification of fresh samples of milk. These samples are acquired from goat, cow, and buffalo milk. A 3-dimensional artificial neural network was proposed for the extraction of spectral features of the milk. A comparison was also made on the proposed methods with that of the existing methods, including the least square discrimination methods and SVM methods. The results prove that the 3D-ANN performs better in terms of F-score, Precision, and Recall than other methods.

Jayanthi et al (2019) introduced a system that used a histogram of the gradients along with the SVM classifier for the detection of fungus in milk. The features of the milk species are captured using real-time cameras and are extracted by the Gradients Algorithm. The features are then fed to the ensemble SVM classification algorithm, which compares with the trained and projects the quality of the samples. The proposed method is implemented using the ARM processor. Experimental results prove that the proposed methods produce very good results in terms of Specificity and Sensitivity.

Mahdi et al (2018) evaluated the portable system that is intelligent in identifying adulteration in milk using the machine-based vision method. The evaluation was done on 13 samples of milk and the mixed ingredients. For the identification of adulteration, the ANNs were deployed and evaluated using a hidden layer with the count of neurons starting from 1 to 20. The best fit network having the structure 12-6-7 is provided for the prediction of the samples with R as the coefficient of correlation and MSE. The proposed methods are found to be effective in terms of Cosine similarity and correlation.

SanaJawaid et al (2017) introduced the SB-ATR-based Fourier transformation as a tool to identify the adulteration in milk. This proposed method is specifically used to test the melanin content in milk. The PLS model was incorporated for identifying the correlation of the data spectrum to melanin concentration. Deep neural networks were deployed to classify pure from impure milk. The proposed technique also took very little time for setup and running time. The performance metrics used here are Entropy, F-Score, Precision, and Recall.

Table 3 shows the gist of research carried out using reinforcement learning for milk adultery identification with the metrics used.

Table 3: Review on reinforcement learning

Author(s)	Method	Advantages	Metrics
Lei-Hong et al	Convolutional neural network	Cross-Validated accuracy of 84%	Root mean square and Average area under the curve
Al-Sarayreh et al	3D convolution neural network (CNN)	spectral-spatial learned features of milk are classified	F score, Precision, Recall
Jayanthi et al	SVM classifier	Real-time capture of fungus growth	Specificity and Sensitivity, Correlation
Mahdi et al(2018)	Artificial Neural Network (ANN)	More samples also yield good results	Cosine similarity and correlation
SanaJawaid et al	Partial least-squares (PLS) model	Very little time complexity	Entropy, F Score, Precision, and Recall

4 Comparative Analysis

The graphical representation shows the comparative analysis of the metrics tabulated in each section and which are obtained from the results of various methods. These graphs are self-

explanatory and are intended to produce a visual comparative analysis. Figure 2 shows the graphical representation of the various results obtained through the methods discussed in Section 3.

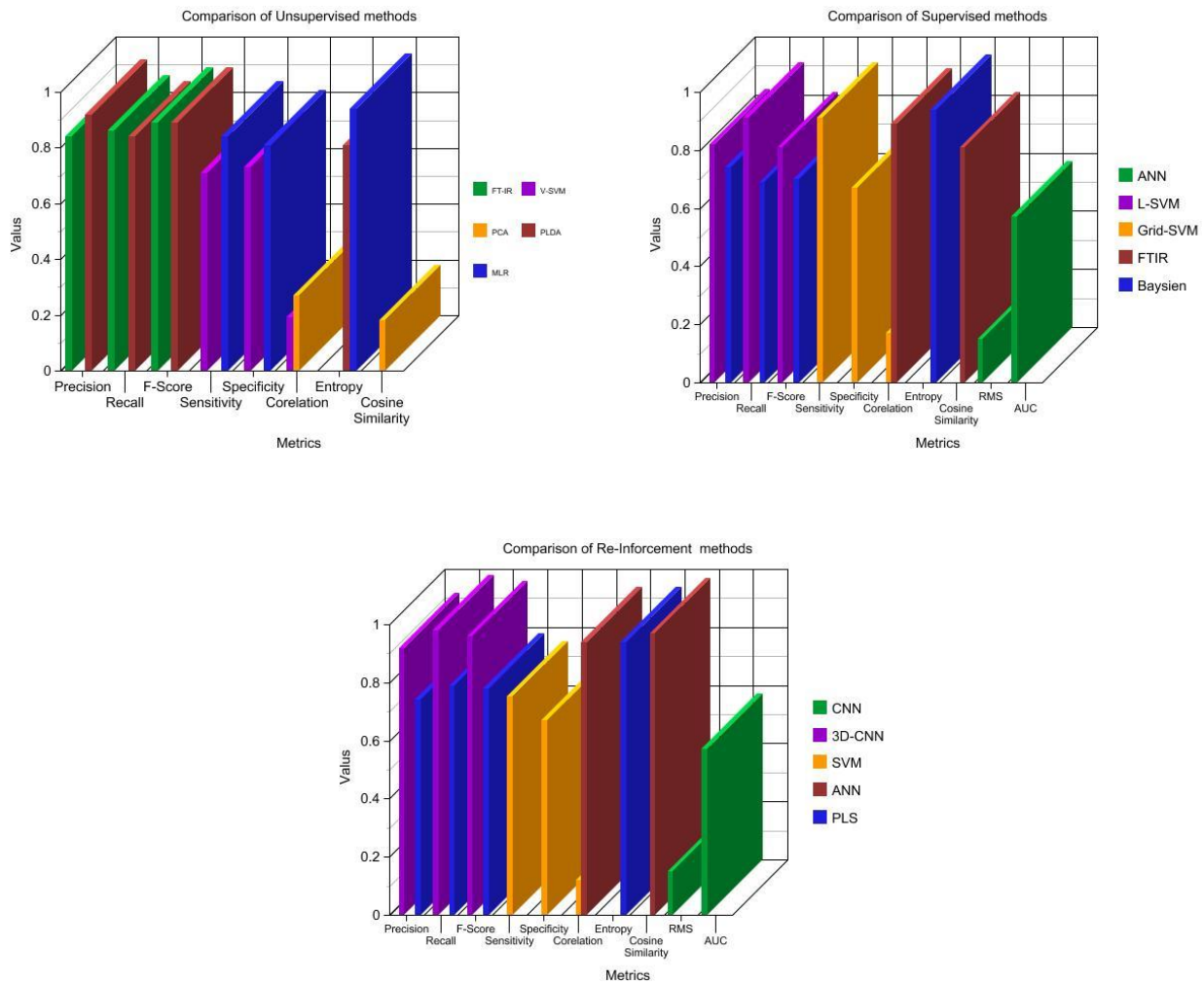


Figure 2: Comparison of unsupervised methods, supervised methods, and reinforcement methods.

5 Conclusion

Food adulteration is the process where the quality is compromised by adding certain chemicals or other substitutions, which, when consumed, cause health hazards. The process includes not only the purposeful addition of additives but also the contamination occurring in the growing stage, storage, and when distributed. Milk is the most common food product being adulterated. Although many studies have been carried out to bring out an accurate method for the detection of adulteration in milk, the usage of Machine Learning in the domain was very limited. This paper gives the gist of such few studies using Machine Learning methods for Milk adulteration detection. This gives an insight into the future research that can be pursued to solve the burning issue of milk adulteration and to save humans from fatal diseases.

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